

On 24 July the World Bank held a stakeholder consultation on 'Lessons Learning and Way Forward for the Housing Reconstruction Programme. The objective of the consultation is to *"understand the concerns and issues of the relevant stakeholders and their views on how to improve and expedite the reconstruction activities. These consultations will assist in preparing for the upcoming Mid Term Review (MTR) of Nepal Earthquake Housing Reconstruction Project (EHRP). The MTR provides stakeholders with the opportunity to review progress, draw learning and, if required, make necessary adjustments in implementation arrangements"*.

In preparation for this, HRRP held a meeting with partners on 19 July¹ to agree the points that HRRP should raise at the stakeholder consultation. The following are the points that were developed and agreed during that session:

Remove the tranche disbursement deadlines, or at least extend for the next two building seasons in rural areas and the next four in urban areas

In late July 2017, the NRA announced deadlines for the disbursement of the housing reconstruction and retrofit grants as follows²:

- Grant agreement signed with local bodies within November 16
- First tranche must be disbursed by January 13, 2018
- Second tranche must be disbursed by April 13, 2018
- Third tranche must be disbursed by July 15, 2018

On 3 April 2018, the NRA Steering Committee agreed to extend the deadline for the disbursement of the second tranche to 16 July 2018³. At the time of writing, this deadline has passed and no decision has been taken on extension as there has been a change in the NRA CEO and a new CEO has yet to be appointed.

The deadlines represent an unrealistic timeframe for reconstruction for many households, and the HRRP advocates for the deadlines to be extended for at least the next two building seasons in rural areas and at least four in urban areas to allow a realistic period for households to complete their reconstruction.

The deadlines are unreasonable in comparison to the timeframe for post-earthquake reconstruction in other contexts and are proving to be counter-productive. They are the main reason that households are constructing small houses just to get the grant, are part of the reason for borrowing at very high rates, and are causing trauma and stress for already stressed people.

In urban areas, the deadlines should ideally be removed, or need to be extended for more than the next four building seasons. Urban reconstruction will take time as it is complex, involves very high levels of investment, and requires a more comprehensive process of collective planning.

In the 18 districts where the grant disbursement process started 15 months after it started in the 14 districts. Households in these districts that have not yet started their reconstruction, often due to confusion about the process and the lack of technical assistance available, will not be able to start and complete their reconstruction within the deadlines. There are also many households in the 18 districts who completed reconstruction of their house early and now find that it is non-compliant and they need to apply corrections to access the second and third tranches of the

¹ [Preparation for World Bank Stakeholder Consultation on Lessons Learned and Way Forward for the Housing Reconstruction Programme, 19 July 2018](#)

² [HRRP Bulletin, 24 July 2017](#)

³ [HRRP Bulletin, 16 April 2018](#)

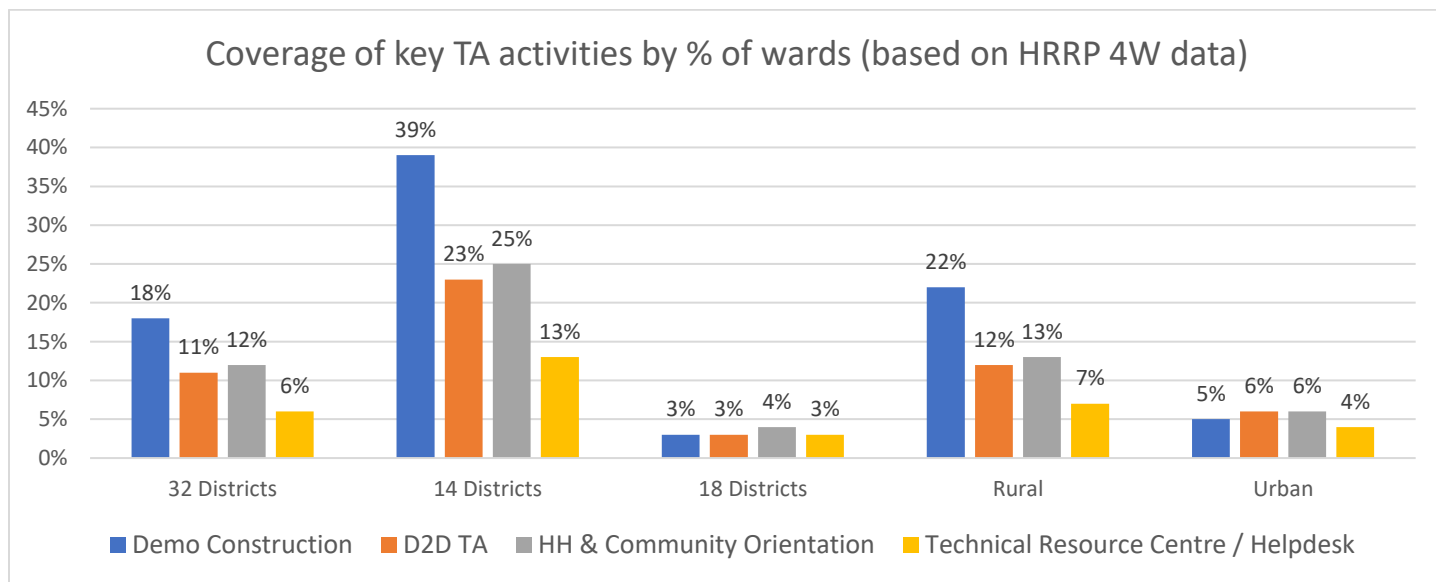
reconstruction grant. Considering the low coverage of technical assistance and high level of non-compliances in these districts, it is important to allow people a reasonable timeframe for reconstruction or to apply corrections.

Unless the deadlines are removed or extended for an appropriate period of time, actions taken to improve the housing reconstruction are unlikely to have any impact.

Scale up coverage of socio-technical assistance, under a collective approach with common standards

It should not be the case that households must rely on luck as to whether they ‘win the recovery lottery’ and get access to socio-technical assistance support. All reconstruction actors have a responsibility to adopt a collective approach with common standards that ensures an equitable level of appropriate support reaches all earthquake affected households. The PDRF states that Strategic Recovery Objective 1 – restore and improve disaster resilient housing, government buildings and cultural heritage, in rural areas and cities – will only be achieved if the financial assistance from the Government of Nepal (GoN) is supported by timely provision of technical assistance, training and facilitation, employing a uniform approach, and with universal coverage across the 32 earthquake affected districts.

To date more than 150 million USD has been spent on technical assistance by Partner Organisations but coverage is low, and there have been issues with quality.



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Where Partner Organisations (POs) are providing comprehensive socio-technical assistance, there is evidence of the very positive impact it has on reconstruction rate and compliance.

The gaps in coverage of socio-technical assistance have been a reality of the housing reconstruction since the beginning. A concerted effort is required from all reconstruction actors to address this for the remainder of the housing reconstruction. It will require investment and collaboration at all levels, amongst all stakeholders.

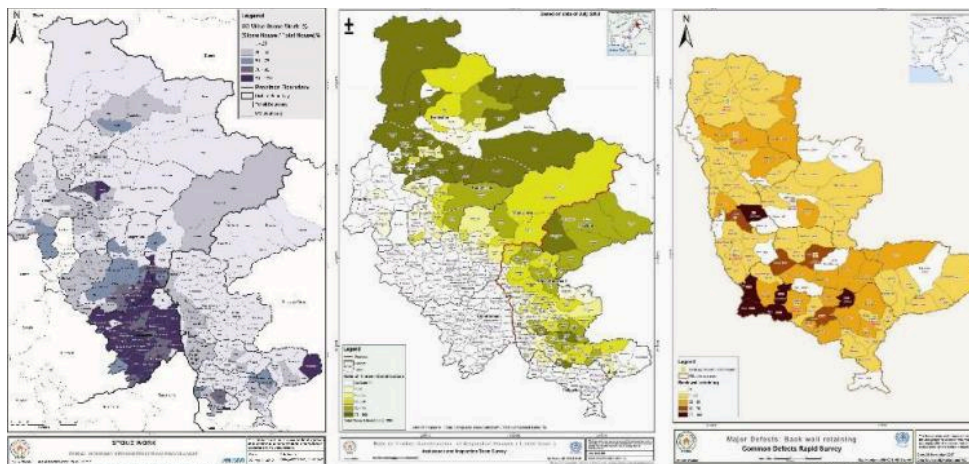
⁴ [Description of Core Socio-Technical Assistance Package, HRRP, August 2017](#)

Based on POs experience to date providing socio-technical assistance, the following is the suggested approach for scale up:

- Mobile masons at household level (3-10 per community)
- Mobile TA teams (one engineer and one social mobiliser minimum), roving at household and local level (1-3 teams per municipality)
- Technical Resource Centres, either at district or hub (covering multiple districts) level
- Coordination and support cell at national level

To date, socio-technical assistance has tended to focus on households that have already started reconstructing. There is a need for socio-technical assistance to be accessible to those that have not started construction, and include information on topics particular to that situation such as budgeting, material quality and purchasing, hiring of labour, designing the house, etc. Socio-technical assistance also needs to be adapted and improved to meet women's needs. Research by HRRP and the Common Feedback Project shows that women are consistently left behind in terms of access to information and socio-technical assistance. If more than 50% of the population are not actively engaged in the reconstruction it is unlikely that it can be a success.

Inspection data holds an enormous amount of valuable data that could contribute to more effective targeting of socio-technical assistance. It can provide detailed information on building typology which allows for targeting of technical information and types of demonstration buildings, information on construction rate which can support with planning inspection visits, scaling up or down of technical assistance activities, and support market monitoring, and it can provide information on quality of construction which supports for targeting of training and technical assistance for particular correction measures and understand the prevalence of non-compliances and can contribute to planning for researching how common non-compliances are coming about. The image below shows examples of how such analysis was used in the housing reconstruction in Pakistan following the 2005 Kashmir Earthquake⁵:



Building Typology : targeting tech information , demo/model buildings

Rate: Progress, targeting training, inspection capacity, market monitoring.

Quality: Prevalence of defects, understanding, targeting training, remedial measures,

⁵ [Presentation on technical assistance in the Kashmir Earthquake housing recovery. Technical Assistance and Training TWG, Shelter Cluster Nepal, 31 May 2015](#)

75 million USD is required for socio-technical assistance to cover gaps and support households to completion of earthquake resilient houses across all 32 earthquake affected districts. This is just 7% of the amount of money yet to be disbursed through the GoN's housing reconstruction grant.

Provide targeted support for vulnerable households and consider seasonal support or support for temporary shelter improvement

The NRA has identified 18,505 households as vulnerable and requiring additional support for their housing recovery. These households were identified by the NRA on the basis of the following vulnerability criteria:

- Senior Citizens above 70
- Single women above 65
- Persons living with disabilities (red and blue card holders)
- Children under 16

HRRP District Coordination Teams (DCTs) have been working with Partner Organisations (POs) and municipalities to verify the vulnerable lists and have so far found that 27% of households on the list have completed construction of their house, 5% have started but have now stopped, 39% have not started, and 29% are in the process of reconstructing their house.

As per the NRA NGO Mobilisation Guidelines, additional support to vulnerable households can include a 'top-up' of up to 50,000 NPRs and should include socio-technical assistance support. Social mobilisation is particularly important for vulnerable support as literacy levels for example can have a big impact on whether or not a household can effectively engage in the process which requires multiple forms to be filled in. Several Partner Organisations (POs) shared their experiences providing support to vulnerable households during a recent HRRP event: <http://hrrpnepal.org/meeting-detail/view/supporting-vulnerable-households-to-reconstruct-learning-from-field-experience->.

Considering that at minimum, 20% of households will take at least another two building seasons to complete their reconstruction, there is a need to consider seasonal support and / or support for temporary shelter improvement in a targeted way. This could include information on rainwater harvesting and drainage that is equally applicable and important for construction of houses.

Vulnerable and marginalised groups need specific, additional support to ensure that they are not left behind in the reconstruction process. This should include those that may not be part of the formal reconstruction programme. Where households may take longer to reconstruct, seasonal support and / or support for temporary shelter improvement should be provided.

Address supply chain constraints and material quality issues, and develop material vendors and producers as agents of technical assistance

Across all districts there are reports of the cost of construction materials going up and up. This is to be expected with the scale of construction work going on. Addressing this requires national level efforts, e.g. cut all tax on cement and steel for 1 year. Transport is another issue that has huge impact on the supply chain. Many remote household face enormous

transportation costs for construction materials. Initiatives looking at supporting collective purchase and / or sharing and supporting transportation costs should be scaled up.

Material quality issues are a concern. Given the limited coverage of socio-technical assistance home owners have had little guidance on how to check material quality and so have limited capacity for strong decision making in this area. For example, many households are constructing with Hollow Concrete Blocks (HCB), but HRRP research conducted in 2016 and 2017 found that just 78% of blocks tested passed the relevant Nepal Standard (NS 119/2042).

Construction material vendors and producers represent fixed points of reference for households and communities, where they can access information and guidance regarding materials and their use for construction. Local technical support should include the mobilisation of vendors and fabricators as partners in the promotion of awareness of the importance of quality assurance in safer construction. It is important to build solidarity and collective responsibility to the community in terms of achieving and maintaining standards. It is also important to have increased knowledge of the construction economy. Some Partner Organisation (POs), including Practical Action and UNDP, have been rolling out this type of support and HRRP recommends that this should be linked with all Technical Resource Centres.

National level action is required to address the increasing costs of construction materials. Need to invest in working with construction material producers and vendors to engage them as technical assistance agents and to address material quality issues.

Mass communications campaign required to address confusion, misinformation, and rumours

There are major issues with confusion, misinformation, and rumours across all earthquake affected areas and all reconstruction stakeholders. Examples include:

- **'Blacklist' and impact of not meeting tranche disbursement deadlines:** many people report fears of missing the tranche disbursement deadlines as they have heard that if the deadlines are missed they will lose their social security payments or their children will not be able to get citizenship.
- **17 designs from DUDBC catalogue volume 1:** the preface of the catalogue states "it is not mandatory for households to select a design from this catalogue, and they are free to prepare house designs outside of the catalogue but these designs must comply with the National Building Code". However, this message has been lost and distorted and many reconstruction actors (NRA, Building DLPIU engineers, POs TA field staff, etc.) have been informing households that they can only reconstruct using one of these designs.
- **One room houses:** in some areas people think that the only way to have an earthquake resistant house, that is eligible for the reconstruction grant, is to build a one room, one storey house. (The deadlines are also causing many people to build one room, one storey houses but this point refers to the misconception that the only earthquake resistant house that can be approved for the grant is a one room, one storey house).

A large-scale mass communications campaign is required to address these rumours and should be flexible to adapt and address other rumours that may arise.

Engaging and working with the media at national and local level will be an essential part of this. This should include regular media engagement sessions at national and local level, and facilitating a space for media to collect accurate information, run fact-checks, etc. with relevant reconstruction actors.

Standardised information to address common rumours and misinformation urgently needs to be widely distributed through a mass communications campaign. Increase engagement with national and local level media.

Develop solutions to address the potential debt crisis we are facing

In October 2017, the HRRP District Coordination Teams (DCTs) spoke by telephone with 570 households who had previously taken part in Inter-Agency Common Feedback Project surveys to collect data on construction costs. They asked three questions:

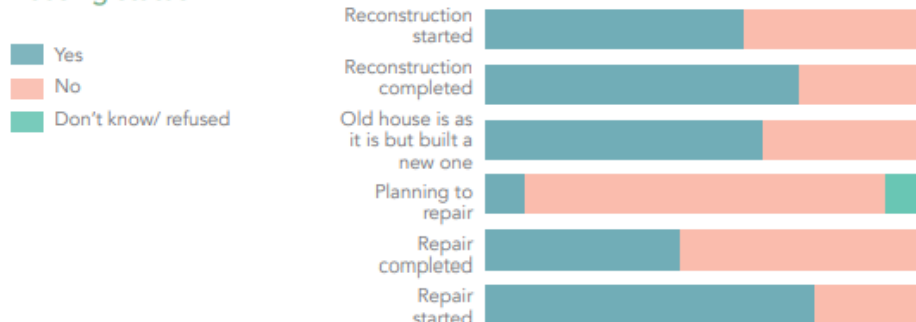
- How much did your house cost / do you estimate it will cost?
- What building materials did you use / are you planning to use and have you salvaged materials?
- Do you plan to / have you already taken a loan and if so for how much, from what type of financial institution, and at what interest rate?

The information provided by the respondents provided indicative data only as it is not a statistically representative sample size. However, it offers useful insight into the financial aspects of reconstruction around which there is a limited amount of existing data.

This data collection⁶ found that the median cost to build a house is 675,000 NPRs (approx. \$6,750), 56% of respondents (321 households) have taken, or are planning to take, a loan to finance the reconstruction of their home, the average interest rate is 23% (annual), and on average the loan amount represents 50% of the total construction costs.

In May 2018, the CFP community perception survey⁷ asked respondents 'have you taken a loan to finance your reconstruction?'. 55% of respondents reported that they have taken a loan and the average interest rate was found to be 24% per annum (almost exactly the same as the HRRP data from October 2017). The difference by housing status was very interesting, with respondents that have not yet started reconstruction much less likely to have taken a loan.

Differences by housing status



⁶ [HRRP Cost of Construction, October 2017](#)

⁷ [Community Perception Report, Common Feedback Project, May 2018](#)

As of 23 July, 807,486 households are eligible for the GoN housing reconstruction grant. If 50% of these households take a loan of 3,250 USD that will be a total of more than 1.3 billion USD in debt. At an average interest rate of 23% per annum, the annual interest payment would be more than 300 million USD. At a household level, if a family is paying interest of 23% per annum, they will pay almost 750 USD in interest annually on a 3,250 USD loan. If the same loan had a 2% interest rate per annum, they will pay 65 USD in interest annually. Over the course of a 5 year loan period (likely a conservative estimate) the different in the amount paid in interest is 3,425 USD (3,750 USD for 23% loan and 325 USD for 2% loan).

Given the scale of the debt, livelihoods programmes intended to improve household income, and in turn allow households to pay off debt, face a possibly insurmountable challenge. It is potentially cheaper, and more effective, for donors / GoN to look at options for clearing debt and / or making the expected subsidised loans a reality.

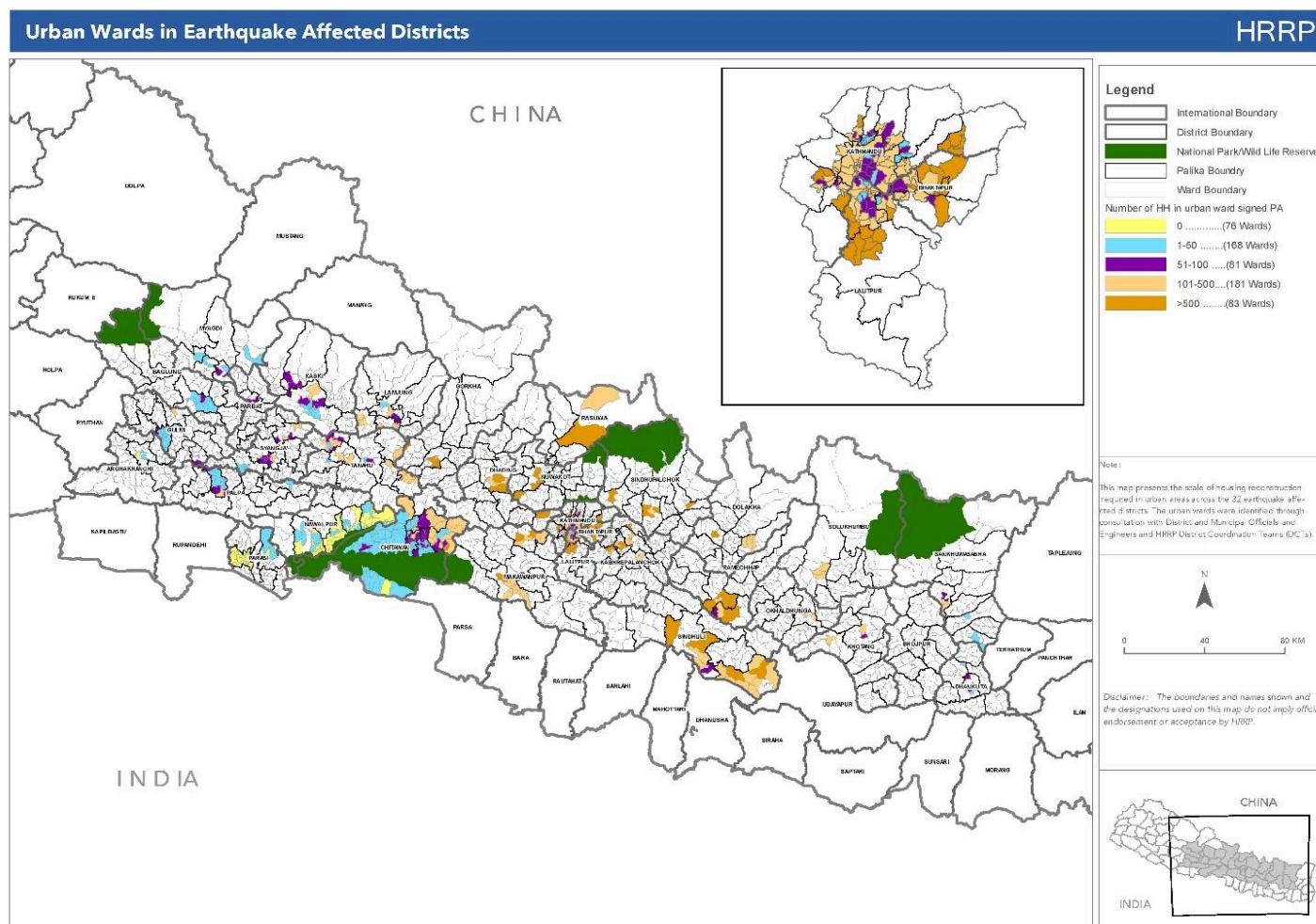
At current rates of borrowing and interest we are facing a huge debt crisis that has potential to negatively impact recovery and living standards in the long term.

Urban housing reconstruction needs investment and targeted policies and approaches

There are urban areas across all 32 earthquake affected districts; 589 urban wards, across 94 municipalities, in total⁸. The map below illustrates the spread of urban wards, as well as the scale of housing reconstruction across each. Ward level data is not available for the total number of households eligible for the housing reconstruction or retrofit grant, but the data for the number of households that have signed the Partnership Agreement (PA) with the GoN is available at ward level. Based on this there are 76 wards where no households have signed a PA, 168 wards where 1-50 households have signed a PA, 81 wards where 51-100 households have signed a PA, 181 wards where 101-500 households have signed a PA, and 83 wards where more than 500 households have signed a PA. Across all urban wards there are more than 120,000 households that have signed a PA.

Defining the urban area in the context of Nepal is challenging as not all wards of Urban Municipalities are actually urban. HRRP has identified the urban wards through consultation with GMALI DLPIU offices, municipal and ward officials, NRA, Building DLPIU technical staff, and the HRRP District Coordination Teams (DCTs). The 589 wards identified are spread across 10 Rural Municipalities, 79 Urban Municipalities, 1 Sub-Metropolitan City, and 4 Metropolitan Cities. Please share any feedback or suggestions on the identified urban wards with Resham Lal Phudhel, HRRP National IM Expert, resham@hrrpnepal.org.

⁸ [HRRP Tool to prioritize additional support for Vulnerable HHs, Version 2, July 2018](#)



Map of urban wards in earthquake affected districts.

The spread of urban wards across earthquake affected districts covers a wide range of areas and contexts and the housing typologies in urban areas reflects this.

There are many elements in the PDRF sector plan that have not yet taken off but that could have a big impact on the urban housing reconstruction. This includes:

- Regeneration of heritage settlements (including 52 in the KTM valley)
- Community grants for local infrastructure – housing recovery is not just about the houses. Settlement planning and recovery are equally important and providing community grants for local infrastructure could contribute hugely to this. ASF Nepal have an interesting project on this for the reconstruction of a 'pati' (rest place) in Bungamati through a local women's group
- Rental housing for urban poor
- Risk sensitive land use plan preparation

A major issue in urban areas is that the high cost of construction means that the GoN housing reconstruction grant represents a much smaller portion of the total house cost than in rural areas. It may be worth considering adjusting policy to allow for a higher grant amount in urban areas. In Gujarat, the government assistance in rural areas was limited

to the maximum grant amount of RS 90,000 (USD 2,117) required to construct a safe core housing unit of 45 square meters but in urban areas, the core unit was estimated to cost RS 175,000 (USD 4,117) for up to 50 square meters⁹.

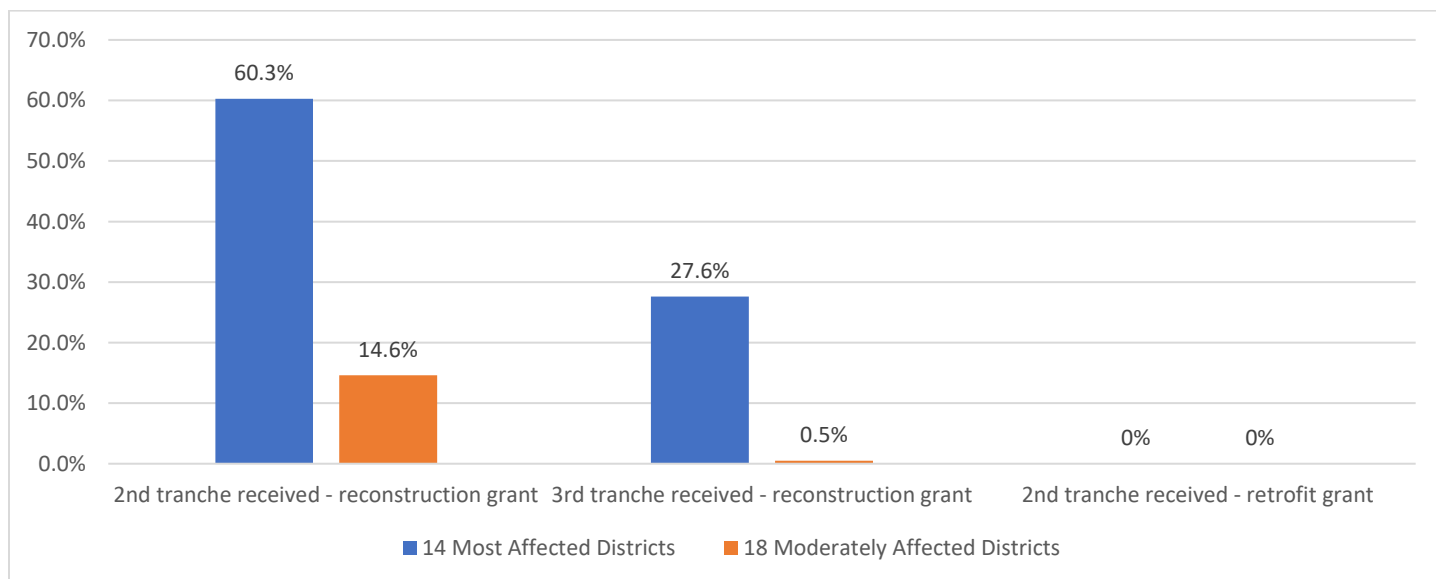
Other major issues in urban areas include debris clearance and demolition of damaged houses, protecting and promoting vernacular architecture, building permit process, urban planning, land issues (right of way, plot sizes smaller than minimum required, eccentric footings, etc.), confusion around retrofitting options, and high level of non-compliances.

Experience from other contexts shows that urban recovery takes longer than rural recovery and requires different policies and support. It is important to factor this in to planning for urban recovery support to make sure that we do not end up in a situation where households in urban areas are reconstructing in 5-10 years with no support available (although there is very limited support available at the moment!).

Urban reconstruction is a complex process, and will require more time than rural reconstruction. Specific policies and incentives for urban reconstruction should be considered.

Increase support to the 18 moderately affected districts

Comparing the grant disbursement data for the 14 most affected districts and the 18 moderately affected districts it is clear that there is a big difference in the progress between the two groups. This is understandable as the grant disbursement process started more than one year later¹⁰ in the 18 moderately affected districts and the coverage of socio-technical assistance has been negligible. However, it is now time to look at ways to better support housing reconstruction in these districts.



⁹ [EAP DRM Knowledge Notes, Working Paper Series No. 9, Housing reconstruction in urban and rural areas](#)

¹⁰ [The Path to Housing Recovery, HRRP, May 2018](#)

NRA, Building DLPIU engineers, sub-engineers, and assistant sub-engineers have been deployed across all 32 earthquake affected districts. They were deployed in the 14 districts in June 2016 and in the 18 districts in December 2017¹¹.

During the HRRP field visits in the 18 districts in February, March, and April of this year, most NRA, Building DLPIU technical staff reported that they had not been trained or oriented on the reconstruction process, policies, and technical guidelines and manuals before being deployed. They are therefore not able to apply these in their work or share relevant information with households and masons. The high level of non-compliant houses and diversity of housing typologies are very challenging for the technical staff to handle with such limited back-stopping and training.

There needs to be increased investment in socio-technical assistance, coordination, data management, and research and development of technical solutions.

For more information refer to the HRRP updated overview of the 18 districts:

https://drive.google.com/open?id=18pSfhZkwC_4ZtKMSiTQV_zD1Cvm0T4ME.

The 18 moderately affected districts have been neglected in terms of investment and support to date. Reconstruction support needs to be spread across all 32 earthquake affected districts moving forward.

Increase support for NRA, Building DLPIU engineers, sub-engineers, and assistant sub-engineers, and District Support Engineers

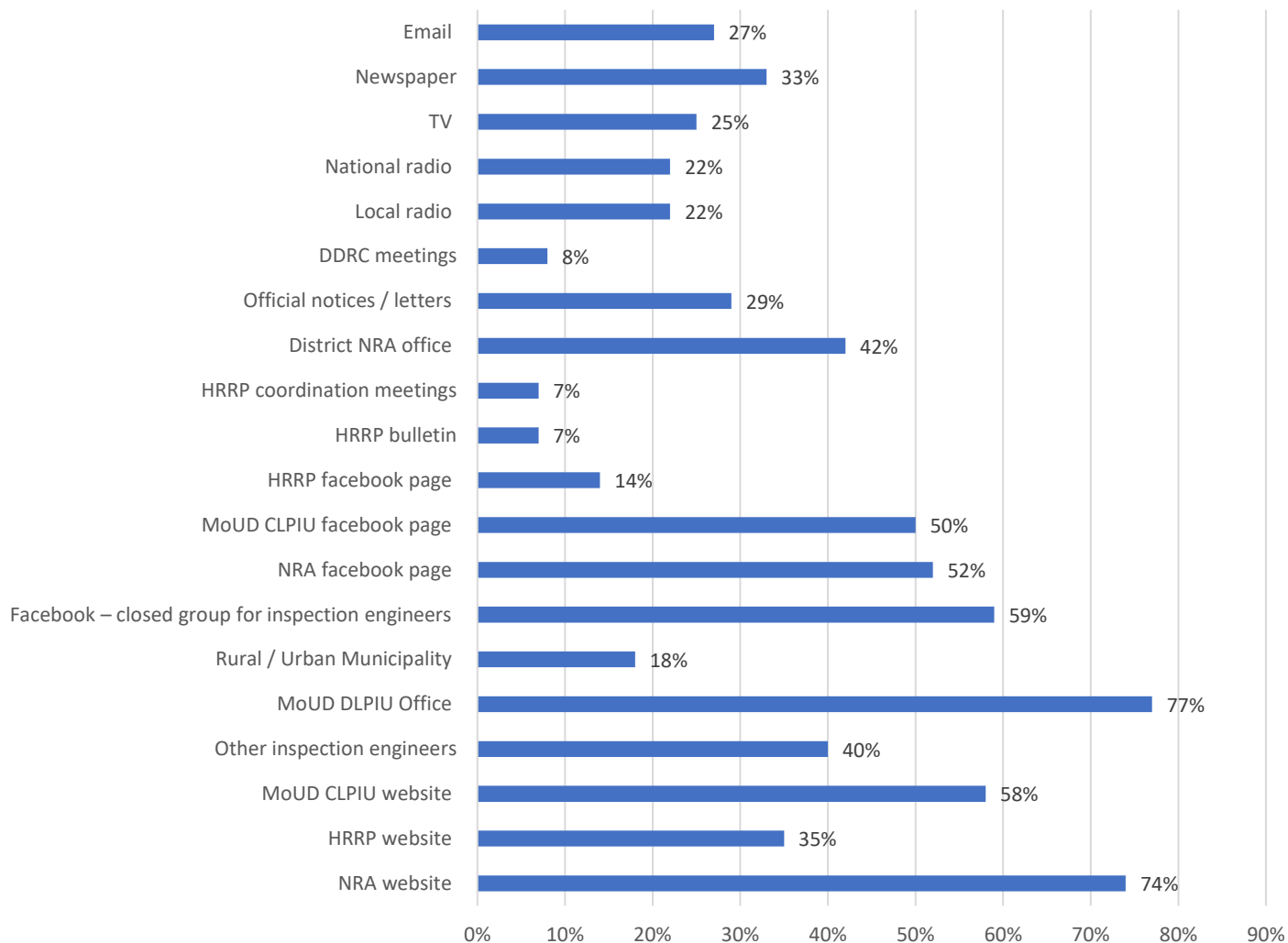
HRRP is currently conducting a survey with NRA, Building DLPIU technical staff to understand how they access information, how they share information with households, gather information on their experience to date, and to provide a space for them to share what support they feel they need.

The survey is not yet complete, but the initial results show the following:

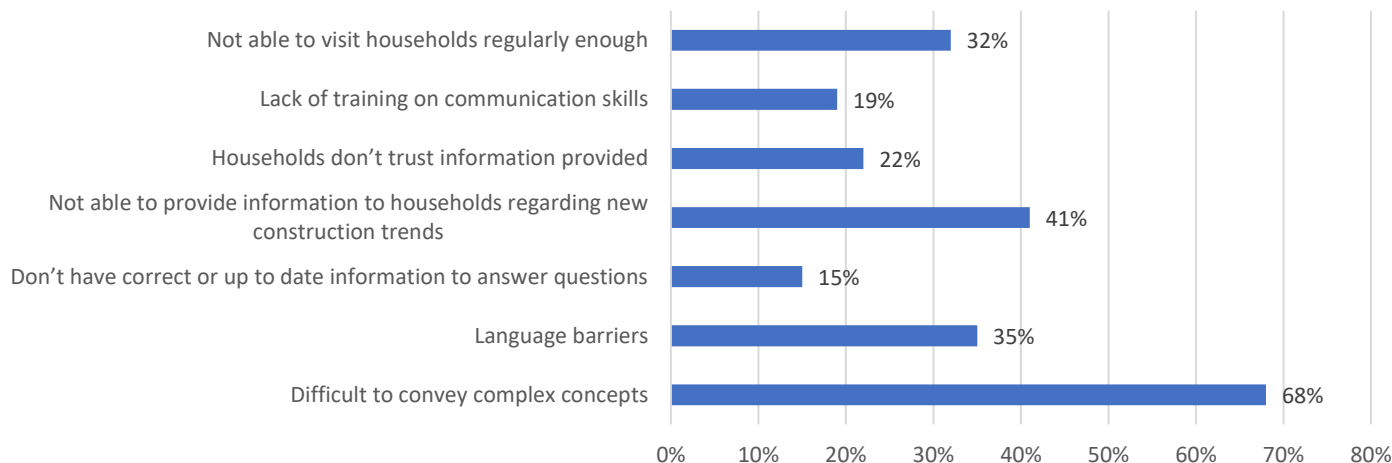
- 47.5% hired in 2016, 31% in 2017, 21.5% in 2018
- 44% did not receive any training before deployment, 24% have not received training since deployment
- Conducting visits to an average of 46 houses per month, average time per visit just under an hour
- Walking is primary mode of transport for 94%
- 61% prepare plan for visits with municipality, 84% based on requests from HHs
- 50% have faced pressure from home owners to pass the house, and 37% have faced pressure from ward / municipal officials
- 36% receive updates from DLPIU once a month, 38% once a week, 11.5% once every two weeks, 14.5% other

¹¹ [The Path to Housing Recovery, HRRP, May 2018](#)

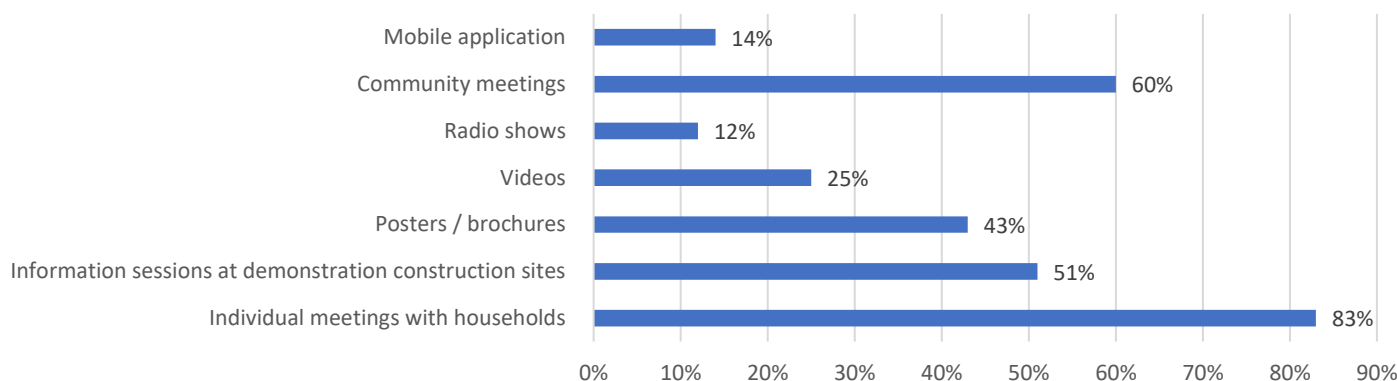
How do you access information on reconstruction standards and policies, and any updates to these?



What are the challenges you face in sharing information with households?



What have you found to be the best ways to share information with households?



There areas where the technical staff feel that additional support is required are as follows:

- Better access to information and improved information sharing processes
- Improved coordination
- Support to address technical issues, e.g. non-compliances
- Partner with local community members / social mobilisers to facilitate work
- Transport, office space

The NRA, Building DLPIU technical staff have an important role to play in the reconstruction in terms of the inspection process, but also as GoN representatives for the reconstruction process at household level and the sole source of technical assistance in many areas. Investment in supporting and developing the technical staff needs to reflect this and consider the impact on longer term technical capacity at local level.

Develop common approach for continuous capacity building support to municipal and ward officials on reconstruction

HRRP is currently conducting a round of data collection in each of the 14 most affected districts to collect information on the capacity building activities that have been conducted with municipal and wards officials to date. Initial analysis of the data collected indicates that there has been a high level of investment to date and there are many actors involved but there are issues with coverage, quality, and standardisation of approach. For example, orientations to municipal officials on the reconstruction policy, procedures, and guidelines have been carried out by a mix of actors each with a different approach, have generally just been one session (not continuous), and there has been limited coverage in the 18 districts. The local officials have a very important role to play in reconstruction but they cannot do this effectively if they are not provided with support and information on a continuous basis under a common approach.

The Ministry of Federal Affairs and General Administration (MoFAGA) has very recently established a unit dedicated to capacity building for local officials. It will be critical to work in collaboration with this unit.

Local officials need continuous capacity building support on the housing reconstruction, under a common approach, if their impact on the reconstruction efforts is to be maximised.

Develop and disseminate design guidelines for alternative / emerging construction technologies and materials, and for traditional construction / architecture

A wide range of construction technologies and materials are being used across the earthquake affected districts – see the [HRRP Flickr](#) for photographs of housing typologies. Some of these are newer construction trends, and some are more traditional. Whilst the National Building Code provides a solid basis for the housing reconstruction, recognising the complexity of updating the whole building code, it is recommended that design guidelines for selection construction technologies and materials are prepared, approved, and disseminated.

Some examples of on-going work in this area include the timber frame and light steel structure manual and the hybrid structure manual which were prepared by the NRA technical standardisation committee and orientations and some trainings on these manuals have been conducted at district level. Manuals are currently being prepared for Hollow Concrete Blocks (HCB) and Compressed Stabilised Earth Blocks (CSEB). An example of an area where research is taking place but the process to prepare a manual / design guideline has yet to start is 2/2.5 storey stone and mud masonry structures.

One of the biggest challenges is not the development of the manuals / design guidelines, but making sure that these get to field level and are utilised. This requires investment in training, printing, and information sharing.

Invest in training, printing, and information sharing to field level to ensure that technical manuals are disseminated and utilised. Support NRA technical standardisation committee with ongoing work to prepare manuals / design guidelines for other construction materials and technologies.

Invest in coordination at all levels

This includes inter-ministerial coordination, coordination between donors, coordination between donors and NRA, coordination between POs, coordination between POs and NRA, etc. and should flow from ward to municipal to provincial to central level. Whilst this is challenging to make a reality, without coordination at all levels there will continue to be the fragmentation, uneven levels of assistance, gaps where there is no assistance, confusing and contradictory advice, inefficient individual efforts to develop tools, and limited information on what is being provided, where and to whom that has been seen to date.

Coordination only works if it is embraced at all levels and leads to an open, collaborative approach amongst all reconstruction actors.

Consider long term institutionalisation of recovery experience now, not just at the end of the reconstruction

The experience from the Nepal housing reconstruction needs to be documented periodically in order to contribute to annual preparedness planning and response in Nepal during monsoon, and to large scale response that may be required for future disasters.

This will require institutionalisation to have impact in the long term. This could be through the new National Disaster Management Agency (NDMA) that is expected to be established in the near future. But it should also be through relevant GoN ministries and departments, donors, and Partner Organisations (POs).

The mid-term review of the EHRP offers a chance to take stock of how recovery experience is / is not being institutionalised and to develop actions to work on this from now, not just at the end of the reconstruction.

Document and institutionalise recovery experience periodically and systematically now – not just at the end of the reconstruction.